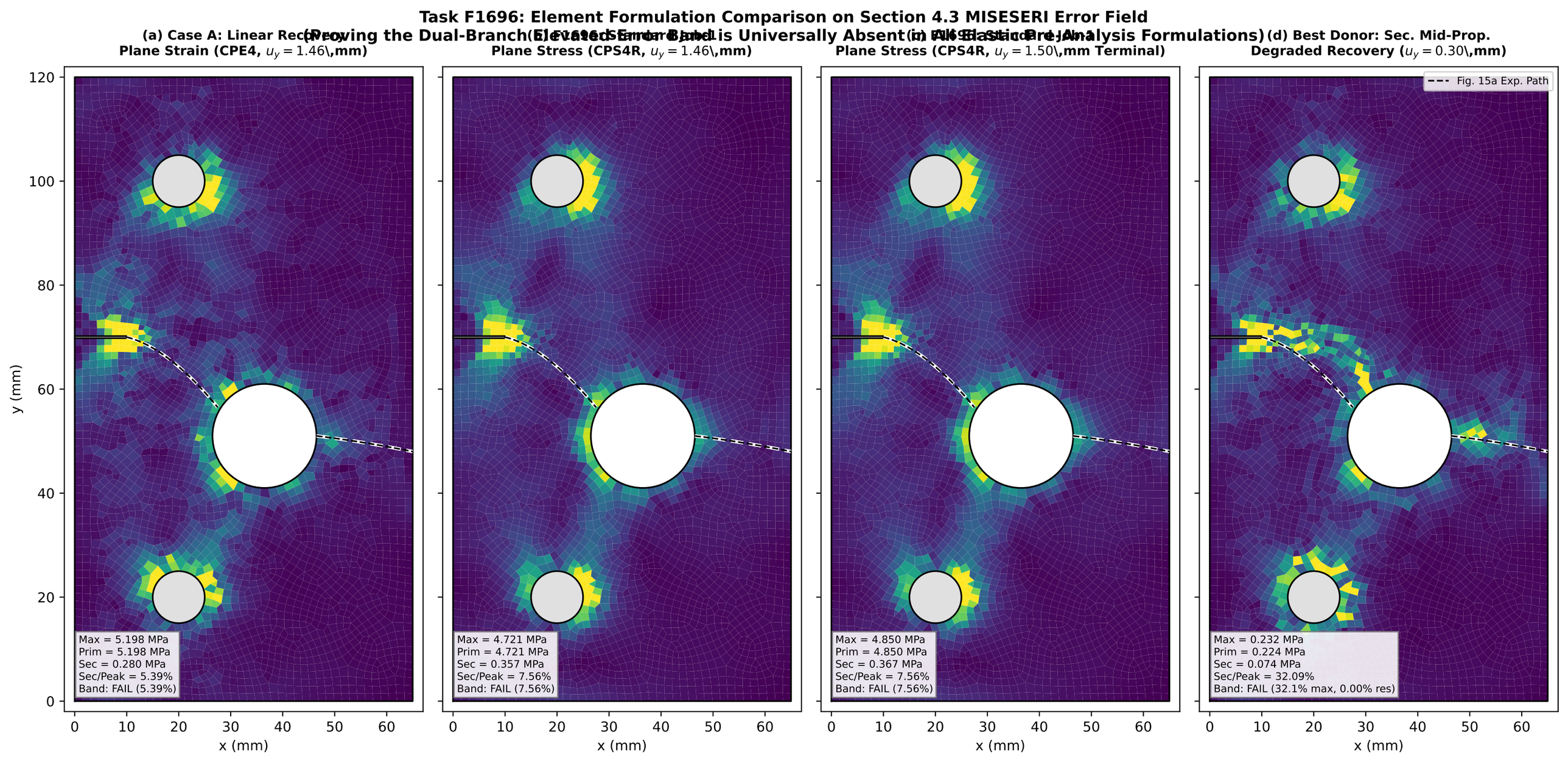

Task F1696: Element Formulation Comparison on Section 4.3 MISESERI Error Field
(Proving the Dual-Branch Element-Based Band is Universally Absent in All Standard Analysis Formulations)
(a) Case A: Linear Recovery, Plane Strain (CPE4, $u_y = 1.46$\,mm)
(b) Case B: Linear Band, Plane Stress (CPS4R, $u_y = 1.46$\,mm)
(c) Case C: Terminal Band, Plane Stress (CPS4R, $u_y = 1.50$\,mm Terminal)
(d) Best Donor: Sec. Mid-Prop. Degraded Recovery ($u_y = 0.30$\,mm)
Fig. 15a Exp. Path
Max = 5.198 MPa
Prim = 5.198 MPa
Sec = 0.280 MPa
Sec/Peak = 5.39%
Band: FAIL (5.39%)
Max = 4.721 MPa
Prim = 4.721 MPa
Sec = 0.357 MPa
Sec/Peak = 7.56%
Band: FAIL (7.56%)
Max = 4.850 MPa
Prim = 4.850 MPa
Sec = 0.367 MPa
Sec/Peak = 7.56%
Band: FAIL (7.56%)
Max = 0.232 MPa
Prim = 0.224 MPa
Sec = 0.074 MPa
Sec/Peak = 32.09%
Band: FAIL (32.1% max, 0.00% res)
y (mm)
x (mm)
0
10
20
30
40
50
60
80
100
120